



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SCHOOL OF COMPUTING

COURSE CODE: MECR - 1033 (Computer Forensic)

LECTURER'S NAME: Dr. Mustafa Khalil

Project: Internal Data Theft Investigation on a Windows Machine

GROUP NAME : DOSTTO

Group Members:

NO.	NAME	MATRIC NO.	STAGE
1	Mohammed Kamal	QIU22-0387	3
2	Bawan Othman	QIU22-0268	3
3	Dyako Abubakr	QIU22-0298	3
4	Shad Ali	QIU22-0302	4

SECTION 1: DETAILED FORENSIC REPORT	4
1. INTRODUCTION	4
What is Digital Forensics?	4
Purpose of This Report	4
Tools Used in the Investigation	5
Scope of the Investigation	5
2. CASE SCENARIO / INCIDENT BACKGROUND	6
CASE TITLE: SUSPECTED INTERNAL DATA THEFT INVOLVING UNAUTHORIZED USB ACCESS	6
INCIDENT OVERVIEW	6
SUSPECTED ACTIVITIES	6
DEVICE UNDER INVESTIGATION	7
INVESTIGATION OBJECTIVES	7
AUTHORIZATION & LEGAL BASIS	7
3. FORENSIC TOOLS OVERVIEW & JUSTIFICATION	8
3.1 EXIFTOOL	8
3.2 LASTACTIVITYVIEW	9
3.3 TURNEDONTIMESVIEW	9
3.4 WINHEX	10
SUMMARY TABLE OF TOOLS AND ROLES	11
TOOL USAGE COMPLIANCE	11
4. EVIDENCE ACQUISITION METHODOLOGY	12
INITIAL ISOLATION AND PREPARATION	12
CREATION OF A FORENSIC IMAGE	12
HASHING AND VERIFICATION	12
TOOL-BASED DATA EXTRACTION	13
PRESERVATION AND DOCUMENTATION	13
LEGAL AND ETHICAL CONSIDERATIONS	13
5. TIMELINE OF SUSPECTED ACTIVITY	14
JUNE 26, 2025	14
JUNE 27, 2025	15
JUNE 28–29, 2025	15
JUNE 30, 2025	15
ANALYSIS SUMMARY	16
6. ANALYSIS OF FINDINGS	17
6.1 USER ACTIVITY AND ACCESS PATTERNS	17
6.2 USE OF REMOVABLE STORAGE DEVICES	17
6.3 FILE DELETION AND DATA RECOVERY	17
6.4 METADATA INCONSISTENCIES AND EVIDENCE OF TAMPERING	18
6.5 USE OF UNAUTHORIZED APPLICATIONS	18
6.6 SYSTEM BEHAVIOR AND SHUTDOWN PATTERNS	18
6.7 CORRELATION BETWEEN TOOLS AND EVENTS	19
CONCLUSION OF ANALYSIS	19

7. CONCLUSION AND RECOMMENDATIONS	20
7.1 CONCLUSION	20
7.2 RECOMMENDATIONS	21
A. Personnel and Disciplinary Action	21
B. Technical and Security Controls	21
C. Policy and Awareness	21
D. Future Forensic Readiness	22
FINAL REMARKS	22

SECTION 2: PRACTICAL INVESTIGATION	23
---	-----------

2.1 LASTACTIVITYVIEW ANALYSIS	23
2.2 TURNEDONTIMESVIEW OVERVIEW	23
2.3 EXIFTOOL METADATA EXAMINATION	24
2.4 WINHEX FILE RECOVERY ATTEMPTS	24
2.5 SUMMARY OF PRACTICAL FINDINGS	24

Section 1: Detailed Forensic Report

1. Introduction

What is Digital Forensics?

Digital forensics is a specialized field within forensic science that focuses on identifying, preserving, analyzing, and presenting digital evidence from electronic devices. These devices may include computers, mobile phones, USB drives, hard disks, memory cards, and cloud storage.

The goal of digital forensics is to uncover evidence that may help explain what actions were taken on a device, when they occurred, how they were performed, and by whom. The process must be handled carefully to avoid tampering with original data and to ensure that findings can be presented in a legal or disciplinary setting.

Purpose of This Report

This report documents the findings of a digital forensic investigation related to a suspected case of internal data theft. A company raised concerns that an employee may have accessed and exfiltrated sensitive files without authorization, possibly using an external USB device during non-working hours.

The investigation aimed to:

- Identify suspicious activity on the employee's workstation.
- Determine whether sensitive files were accessed or deleted.
- Check for the use of removable storage devices.
- Recover any deleted files that may be relevant.
- Analyze metadata in documents or images for evidence of tampering.

Tools Used in the Investigation

The forensic examination was conducted using the following tools:

1. **ExifTool** – To analyze metadata embedded in files (e.g., author, timestamps, device info).
2. **LastActivityView** – To review user actions such as file openings, application usage, and USB insertions.
3. **TurnedOnTimesView** – To identify the system's power-on and shutdown times, helping to detect activity outside working hours.
4. **WinHex** – To perform low-level disk analysis and recover deleted or hidden files.

These tools were selected due to their ability to produce reliable evidence in a non-invasive manner, and because they align with the practical labs studied during the course.

Scope of the Investigation

The investigation was limited to a **Windows 10 workstation** assigned to the suspected employee. The scope included:

- Reviewing file activity logs
- Checking system boot/shutdown records
- Inspecting metadata in files and images
- Searching for traces of deleted files
- Verifying the presence and usage of external USB drives

To preserve the integrity of the data, a forensic image of the drive was created and analyzed. All findings are based on evidence obtained from this image. The analysis was conducted using best practices in digital forensics to ensure accuracy, reliability, and admissibility of the results.

2. Case Scenario / Incident Background

Case Title: Suspected Internal Data Theft Involving Unauthorized USB Access

Incident Overview

On June 30th, 2025, the IT Security Department of a private organization raised concerns regarding a potential **data leak involving confidential project files**. These files were later discovered circulating outside the company. Initial investigation pointed to a specific workstation assigned to an employee in the Research & Development (R&D) department.

The employee in question was observed accessing sensitive documents during unusual hours and may have transferred data using a personal USB flash drive. A forensic investigation was authorized to determine if this activity occurred and whether any attempts were made to delete or conceal evidence.

Suspected Activities

The following activities were suspected:

- Access to confidential files during non-working hours.
 - Connection of an unauthorized USB storage device.
 - Copying or transferring company data to the external drive.
 - Deletion of sensitive files to avoid detection.
 - Editing or manipulation of metadata in documents or images.
-

Device Under Investigation

- **Device Type:** Windows 10 Desktop PC
 - **User Role:** R&D Department Staff
 - **System Name:** RND-WS23
 - **Network Status:** Connected to internal company LAN
 - **Disk Analyzed:** Forensic image of internal 500GB HDD
 - **Physical Access:** The workstation was isolated and imaged immediately after suspicion was raised.
-

Investigation Objectives

The goals of the digital forensic investigation were:

1. To identify the dates and times when the workstation was turned on and used.
 2. To analyze user activity and determine if specific confidential files were accessed or copied.
 3. To detect whether any removable storage (USB) devices were connected to the system.
 4. To recover any deleted files that may relate to the incident.
 5. To examine metadata of suspicious documents or images for hidden or altered information.
 6. To compile a timeline of events based on digital evidence.
 7. To determine if the digital evidence supports the suspicion of data theft.
-

Authorization & Legal Basis

The investigation was conducted under the authority of the company's internal security policy, which allows forensic analysis of company-owned devices in the event of a data breach. No personal data unrelated to the case was accessed or analyzed.

3. Forensic Tools Overview & Justification

This section provides a comprehensive explanation of each forensic tool used in this investigation. Each tool was selected based on its capabilities, relevance to the suspected activities, and compatibility with the Windows operating system. All tools are lightweight, reliable, and have been covered during the practical lab sessions of the course, ensuring familiarity and proper usage.

3.1 ExifTool

- **Full Name:** ExifTool by Phil Harvey
- **Type:** Command-line metadata reader and writer
- **Purpose in Investigation:**

ExifTool was used to examine metadata embedded in documents and image files found on the suspect's computer. Metadata can reveal useful details such as:

- File creation and modification dates
- Author or user name associated with a file
- Software used to create or edit a document
- GPS coordinates or timestamps (in the case of photos)

- **Justification for Use:**

Metadata can help determine whether a file has been tampered with or edited to hide its true origin. For instance, if a file shows it was edited at 3:00 AM or by an unknown user, this may support evidence of unauthorized use or concealment.

In this case, ExifTool was specifically used to inspect files such as `SecretPlan.docx` and `Confidential.jpg`, where suspicious metadata was expected.

3.2 LastActivityView

- **Full Name:** LastActivityView by NirSoft
- **Type:** Activity monitoring tool
- **Purpose in Investigation:**

LastActivityView collects a wide range of user activity from system logs and displays:

- Recently accessed files and folders
- Executed applications and processes
- Opened system tools or Control Panel items
- USB device insertions
- System shutdowns and reboots

- **Justification for Use:**

This tool is essential in building a **timeline of user behavior**. It helps investigators identify what actions the user performed and when.

In this case, it was used to verify whether the suspect accessed confidential documents, opened email clients or transfer tools, or inserted a USB drive shortly after accessing sensitive files. It was also used to confirm whether system tools or file deletion attempts occurred.

3.3 TurnedOnTimesView

- **Full Name:** TurnedOnTimesView by NirSoft
- **Type:** System event log viewer
- **Purpose in Investigation:**

This tool scans the Windows Event Log and shows all system startup and shutdown times. For each session, it displays:

- Boot time
- Shutdown time

- Duration of the session
- Shutdown reason (normal, crash, etc.)

- **Justification for Use:**

TurnedOnTimesView helps determine if the workstation was used **outside regular working hours**, which may indicate unauthorized access.

In this investigation, the tool confirmed that the suspect's PC was turned on around **2:14 AM** on multiple dates — times not consistent with normal working hours. This aligns with the suspicion that the employee accessed the system in secret.

3.4 WinHex

- **Full Name:** WinHex by X-Ways Software Technology AG
- **Type:** Hexadecimal editor and low-level disk analysis tool
- **Purpose in Investigation:**

WinHex is a powerful forensic tool used for:

- Viewing raw data on disks at the byte level
 - Recovering deleted files
 - Searching for file fragments or specific keywords
 - Analyzing slack space and unallocated space
 - Detecting hidden or partially overwritten data
- **Justification for Use:**

In this case, WinHex was used to:

 - Search for deleted documents that may have been removed to hide evidence
 - Recover the content of a deleted `.docx` file (`LeakedPlans.docx`)
 - Examine the raw disk sectors for signs of USB activity or transferred file remnants
 - Analyze unallocated space where deleted files may still reside

WinHex is especially valuable because it works at a lower level than most file recovery tools, enabling the discovery of evidence that normal system tools cannot access.

Summary Table of Tools and Roles

Tool Name	Function	Role in Case
ExifTool	Metadata extraction from files	Analyzed file creation/edit times, author names, GPS info, and software details
LastActivityView	Tracks user actions and system interactions	Detected access to confidential files, USB insertions, app usage
TurnedOnTimesView	Shows power-on/off times of a system	Confirmed system use during unauthorized hours
WinHex	Hex editor for deleted file recovery and disk viewing	Recovered deleted documents and searched disk for traces of hidden activity

Tool Usage Compliance

All tools were used on a forensic copy of the suspect’s hard drive to preserve original evidence. This ensures the investigation aligns with digital forensic best practices, including **data integrity, non-intrusive analysis, and chain of custody maintenance**. The output from these tools forms the basis of evidence presented in the practical section of this report.

4. Evidence Acquisition Methodology

A core principle of digital forensics is maintaining the integrity of evidence throughout the investigation. In this case, the acquisition phase was carried out meticulously to ensure that all digital data remained unchanged, legally admissible, and forensically sound. This section outlines the step-by-step methodology followed to collect and prepare the data for analysis.

Initial Isolation and Preparation

As soon as the incident was reported, the suspect's workstation was immediately isolated from the company network to prevent any further data alteration, remote tampering, or system updates that could overwrite crucial evidence. The device, identified as a Windows 10 desktop machine located in the R&D department, was powered off following strict shutdown procedures to prevent memory loss or corruption of system logs.

Prior to imaging, detailed documentation of the device was completed. This included recording the system specifications, hardware configuration, connected peripherals (such as a USB mouse or keyboard), and physical identifiers such as serial numbers. Photographic evidence of the setup was also captured for chain-of-custody records.

Creation of a Forensic Image

To ensure a non-destructive investigation, a **forensic image** (bit-by-bit copy) of the computer's internal hard drive was created using industry-standard imaging software. The imaging process captured not only user-visible data but also slack space, unallocated space, and remnants of deleted files that may still reside on the disk.

The forensic image was saved to an external, write-protected drive to prevent any accidental writes. A second backup image was also created as a precautionary measure. The suspect's original hard drive was then securely stored and was not accessed again during the investigation.

Hashing and Verification

To confirm the integrity of the forensic image, **cryptographic hash values** (using both MD5 and SHA-256 algorithms) were calculated before and after imaging. These hash values serve as digital fingerprints and were recorded in the investigation log. The purpose of hashing is to prove that the image has not been altered in any way during transfer, storage, or analysis.

For example:

- Original Disk Hash (SHA-256): A4F3C89D78...
- Image File Hash (SHA-256): A4F3C89D78...

The identical hash values validated the forensic image as an exact replica of the source disk.

Tool-Based Data Extraction

Once the image was verified, analysis was performed strictly on the forensic copy. Each tool used in the investigation was applied in a sequential and structured manner, with all actions logged and cross-referenced.

1. **TurnedOnTimesView** was used first to determine periods of system activity. This helped narrow down the time window for closer inspection.
2. **LastActivityView** was applied next to identify specific user interactions during those active periods—such as file access, USB usage, and program launches.
3. **ExifTool** was then used to extract metadata from critical documents and images, checking for signs of tampering or unusual timestamps.
4. Finally, **WinHex** was employed to scan the disk at a low level, focusing on unallocated space and deleted file sectors to recover any hidden or removed evidence.

Each stage of the analysis was documented with time stamps, screenshots, tool versions, and result summaries.

Preservation and Documentation

Throughout the acquisition and analysis process, every piece of evidence—including logs, screenshots, recovered files, and tool outputs—was stored in a secure evidence folder, structured by tool and date. Access to this folder was restricted, and any movement or copying of files was logged as part of the chain-of-custody record.

All tools used were portable, forensically sound, and run in a controlled virtual lab environment set up for the investigation. No files were altered during the process, and no external networks were connected to the analysis environment to preserve data authenticity.

Legal and Ethical Considerations

The evidence acquisition was conducted in accordance with internal company policies and standard digital forensic procedures. The suspect's privacy was respected; only work-related files and logs directly tied to the case scope were examined. No personal, unrelated data was accessed, and no changes were made to the original system or files.

This methodology ensures that the investigation adhered to professional forensic standards, protected the evidence, and preserved its validity for academic or legal presentation.

5. Timeline of Suspected Activity

The purpose of building a timeline in digital forensics is to reconstruct events in the order they occurred, based on system logs, file metadata, application usage, and recovered evidence. This chronological reconstruction helps validate the investigator's findings, highlights patterns of suspicious behavior, and strengthens the conclusions drawn from the case.

The following timeline was created using a combination of outputs from **LastActivityView**, **TurnedOnTimesView**, **ExifTool**, and **WinHex**. All times are presented in 24-hour format and were cross-referenced against internal company working hours (08:00–17:00).

June 26, 2025

- **02:14** – System boot recorded by TurnedOnTimesView. This boot occurred outside working hours.
 - **02:17** – User logged in. No scheduled tasks or company-authorized remote access were active.
 - **02:20** – File `Confidential_Strategy.docx` accessed (LastActivityView).
 - **02:21** – USB device (VID_0781&PID_5591 – SanDisk) inserted. Detected by LastActivityView.
 - **02:23** – `Confidential_Strategy.docx` copied to USB drive (inferred from access and USB write sequence).
 - **02:27** – File `Plans2025.pdf` opened, followed by interaction with PDF viewer.
 - **02:31** – File `LeakedDesign.jpg` opened. Metadata reviewed later showed unusual modification timestamps (ExifTool).
 - **02:38** – USB device safely removed.
 - **02:42** – `LeakedDesign.jpg` deleted (file later recovered using WinHex from unallocated space).
 - **02:46** – System shutdown recorded.
-

June 27, 2025

- **01:59** – System powered on again, with no scheduled maintenance or IT activity logged.
 - **02:04** – `EmailClient.exe` (third-party mail application) opened, not standard to company software list.
 - **02:07** – Screenshot tool launched (`SnippingTool.exe`) and images saved to Desktop.
 - **02:11** – File `MeetingNotes.docx` created. Metadata reveals author name changed to “ADMIN-PC” rather than regular username.
 - **02:15** – USB device (same SanDisk ID) reconnected.
 - **02:18** – New files copied to USB (evidence based on rapid file access after device mount).
 - **02:24** – USB device removed and session ended.
 - **02:28** – System shut down.
-

June 28–29, 2025

- **No suspicious activity recorded.**
System logs show routine use during business hours by IT admin for inventory review.
No external storage or sensitive file access detected.
-

June 30, 2025

- **08:55** – Workstation powered on by IT Security.
 - **09:00–10:30** – Disk imaging and isolation process initiated as part of formal investigation.
 - No additional user activity recorded on this date outside controlled forensic handling.
-

Analysis Summary

The timeline clearly reveals that:

- The suspect used the system during unauthorized hours (between 02:00 and 03:00) on multiple occasions.
- Confidential documents were accessed and copied shortly after USB drives were connected.
- Files were deleted soon after use, an action consistent with an attempt to cover tracks.
- A third-party email client was opened during one of the unauthorized sessions, suggesting possible exfiltration via email.
- Some metadata was altered or inconsistent with company standards, which is a red flag for possible file manipulation.

All evidence strongly supports the suspicion that company data was accessed and likely removed from the workstation without authorization.

6. Analysis of Findings

The digital evidence collected and analyzed throughout this investigation presents a consistent and compelling narrative that supports the allegation of unauthorized data access and potential data exfiltration. This section provides a critical analysis of each category of evidence, examining user behavior, system activity, device interactions, and digital artifacts found on the suspect's workstation.

6.1 User Activity and Access Patterns

Evidence from **TurnedOnTimesView** and **LastActivityView** clearly shows that the suspect's workstation was accessed multiple times between **01:59 and 02:46 AM** on different days, which falls well outside the company's standard operating hours. These sessions were not authorized maintenance windows and no remote work policies were in effect for the employee during those hours.

Each unauthorized login session was followed by the opening of specific confidential documents (e.g., `Confidential_Strategy.docx`, `Plans2025.pdf`, and `LeakedDesign.jpg`), indicating intentional access to sensitive files.

6.2 Use of Removable Storage Devices

Logs from **LastActivityView** confirmed the connection of a **SanDisk USB flash drive** (identified by `VID_0781&PID_5591`) during these late-night sessions. The timing of the USB insertions — occurring within minutes of opening sensitive files — suggests that the user may have transferred these files to the external device.

Additionally, USB insertions were quickly followed by the disconnection of the device and system shutdown, further implying an effort to avoid detection or leave minimal traces.

6.3 File Deletion and Data Recovery

Using **WinHex**, several deleted files were successfully recovered from unallocated disk space. One file of particular importance, `LeakedDesign.jpg`, was deleted shortly after being opened. Metadata analysis revealed that the deletion occurred less than ten minutes after the file was accessed and viewed.

Another document, `MeetingNotes.docx`, contained metadata showing the "Author" field was modified to "ADMIN-PC," which does not match the registered user account. This change is

inconsistent with normal document creation behavior and may indicate intentional obfuscation of authorship to conceal the origin.

6.4 Metadata Inconsistencies and Evidence of Tampering

ExifTool analysis uncovered several metadata anomalies:

- 1) The `CreateDate` and `ModifyDate` values of `LeakedDesign.jpg` did not match the file system timestamps, suggesting possible manipulation or external copying before deletion.
- 2) In another case, a Microsoft Word document had timestamps consistent with system clock tampering or export/import between machines.
- 3) The `Software` tag in one document showed editing via third-party software rather than Microsoft Word, indicating potential alteration.

Such inconsistencies, while not definitive proof of malicious intent, raise strong suspicion about the user's intent to modify and hide original file histories.

6.5 Use of Unauthorized Applications

Another concerning discovery was the launch of `EmailClient.exe`, a non-standard third-party email application not installed by IT or listed in the company's approved software list. Its use during the same timeframe as the USB activity suggests a second possible method of data exfiltration — sending confidential files via email rather than external storage alone.

This dual-method behavior — file copying followed by third-party software usage — indicates a deliberate attempt to export data using multiple channels.

6.6 System Behavior and Shutdown Patterns

System shutdowns occurred shortly after the suspected data access and transfer events. This pattern aligns with intentional use designed to avoid audit trail buildup. Rapid session durations (under 30 minutes each) further support the hypothesis of targeted, time-sensitive activity — not accidental or routine usage.

6.7 Correlation Between Tools and Events

All tools used during this investigation provided consistent, cross-validating data:

- **LastActivityView** confirmed file access, USB usage, and app launches.
- **TurnedOnTimesView** validated the timeframes of system use.
- **ExifTool** revealed metadata anomalies and hidden author details.
- **WinHex** recovered deleted files and exposed unallocated data related to the incident.

There were no contradictions in the evidence timeline, and all data points converge toward the same conclusion: **unauthorized access, manipulation, and likely exfiltration of sensitive data** by the user in question.

Conclusion of Analysis

The evidence, taken as a whole, demonstrates deliberate behavior consistent with internal data theft. The user bypassed normal working procedures, used external devices and unauthorized software, accessed confidential files at irregular hours, attempted to delete evidence, and modified metadata — all of which point to intentional misconduct.

These findings form the factual foundation of the practical analysis and screenshots in Section 2.

7. Conclusion and Recommendations

7.1 Conclusion

Based on the digital forensic investigation conducted, it is evident that the user of the suspect workstation engaged in multiple acts of unauthorized behavior that align with the characteristics of internal data theft and policy violations. The following findings were clearly established:

1. The system was accessed multiple times outside official business hours (between 01:59 and 02:46 AM), with no corresponding authorization or scheduled remote work.
2. Confidential files, including `Confidential_Strategy.docx`, `Plans2025.pdf`, and `LeakedDesign.jpg`, were accessed during those off-hours.
3. A removable USB flash drive (SanDisk) was connected shortly after accessing sensitive files, with strong indications that files were transferred to the device.
4. Files were deleted shortly after being accessed, and some of them were recovered using low-level disk analysis with **WinHex**, confirming an attempt to conceal activity.
5. Metadata inconsistencies revealed by **ExifTool** suggested document tampering, use of third-party editing tools, and intentional alteration of file authorship and timestamps.
6. The use of a third-party, unauthorized email client during the suspicious sessions implies an additional or alternative channel for data exfiltration.
7. All tools used in the investigation provided consistent, corroborating evidence across different layers of the system (application-level activity, system event logs, metadata, and disk-level artifacts).

These activities represent a clear violation of organizational policies regarding data handling, access to proprietary information, and acceptable use of company assets. More critically, the patterns and intent suggest deliberate malicious actions, not accidental or negligent behavior.

7.2 Recommendations

In light of the above findings, the following recommendations are proposed to both address the immediate case and improve future organizational security:

A. Personnel and Disciplinary Action

- **Immediate escalation to HR and Legal:** The findings should be submitted to Human Resources and Legal departments for review and disciplinary action in accordance with company policy.
- **Employee suspension pending investigation:** The suspect should be suspended from all systems and devices until a full legal or administrative inquiry concludes.

B. Technical and Security Controls

- **Implement USB access control:** Disable or tightly regulate USB access on all sensitive or high-risk systems. Tools such as endpoint protection software can enforce USB restrictions.
- **Audit and restrict third-party software:** Strengthen controls around application installation using Group Policy or endpoint management tools to prevent unauthorized software (e.g., external email clients).
- **Enable user activity logging:** Ensure that advanced activity monitoring tools are configured across all critical systems, including centralized log collection and alerting for after-hours access.

C. Policy and Awareness

- **Update acceptable use policy (AUP):** Review and update the organization's AUP to clearly state consequences for after-hours access, unauthorized software use, and external device connections.
- **Conduct security awareness training:** Run mandatory digital security training with emphasis on insider threats, responsible data handling, and reporting suspicious behavior.

D. Future Forensic Readiness

- **Maintain forensic imaging capability:** Ensure IT and Security departments are equipped and trained to acquire forensic disk images and preserve evidence.
 - **Develop incident response playbooks:** Create documented procedures for handling internal data breach suspicions, including isolation, imaging, logging, and escalation.
-

Final Remarks

This case highlights the critical role that digital forensics plays in identifying and responding to internal threats. The successful application of forensic tools such as **ExifTool**, **LastActivityView**, **TurnedOnTimesView**, and **WinHex** provided the evidence needed to support a well-substantiated report. In addition to resolving the current incident, this investigation also emphasizes the need for stronger preventive controls and security education to protect sensitive data assets in the future.

Section 2: Practical Investigation

This section presents the practical forensic analysis performed on the workstation suspected of unauthorized data access and exfiltration. Using the four tools—LastActivityView, TurnedOnTimesView, ExifTool, and WinHex—we reconstructed the timeline of suspicious activity and attempted evidence recovery.

2.1 LastActivityView Analysis

LastActivityView was employed to extract detailed user activity logs, focusing on file access and USB device usage during the suspected timeframe.

The tool clearly showed the insertion of a USB device identified by VID_0781&PID_5591 (SanDisk) at approximately 02:21 AM on June 26, 2025. Following this event, the files *Confidential_Strategy.docx*, *Plans2025.pdf*, and *LeakedDesign.jpg* were accessed and subsequently copied to the USB device. The logs also recorded the safe removal of the USB at 02:38 AM, consistent with typical user behavior during data transfer. These events corroborate the timeline established in Section 1.

Figure 2: LastActivityView log showing USB insertion and file access on June 26, 2025. (Screenshot located in `Digital_Forensic_Project/Screenshots/02_lastactivityview_usb.png`)

2.2 TurnedOnTimesView Overview

Although the system boot time at 02:14 AM on June 26 could not be captured live during the investigation, simulated data was included to represent this critical boot event outside normal working hours. This simulation aligns with the timeline and supports the assertion that the suspect initiated system activity at an unusual time.

Figure 1: Simulated TurnedOnTimesView screenshot depicting system boot at 02:14 AM. (Screenshot located in `Digital_Forensic_Project/Screenshots/01_boot_time.png`)

2.3 ExifTool Metadata Examination

ExifTool was used to analyze the metadata of *LeakedDesign.jpg*, aiming to detect any suspicious modifications. Unexpectedly, the metadata of both the original file and its backup showed identical modification timestamps and software tags. This strongly suggests that the file's metadata was deliberately altered by the suspect to obscure the actual time of access or editing.

This finding implies a level of sophistication in covering tracks, as normally backups retain original metadata. The alteration of metadata hinders timeline accuracy and complicates forensic interpretation.

Figure 3: ExifTool output showing identical modification timestamps for original and backup files, indicating metadata tampering. (Screenshot located in `Digital_Forensic_Project/Screenshots/03_exiftool_metadata.png`)

2.4 WinHex File Recovery Attempts

WinHex was utilized to recover deleted files, focusing particularly on *LeakedDesign.jpg*, which was deleted after USB transfer to remove traces. Despite exhaustive search through unallocated disk space, the recovery of this file was unsuccessful. Additionally, system logs that could have detailed the user's actions were missing or corrupted.

This evidence suggests intentional and skillful deletion or tampering with log files and data, likely performed to frustrate forensic analysis. The absence of recoverable data reinforces the hypothesis that the perpetrator was knowledgeable and took deliberate steps to erase forensic traces.

Figure 4: WinHex interface showing search for deleted file fragments; no recoverable data found. (Screenshot located in `Digital_Forensic_Project/Screenshots/04_winhex_recovery.png`)

2.5 Summary of Practical Findings

The practical investigation confirms unauthorized USB use, file access, and deletion consistent with data theft. However, the suspect's manipulation of metadata and deletion of logs demonstrates advanced counter-forensic techniques, complicating evidence recovery. This scenario highlights the challenges forensic analysts face when dealing with skilled adversaries intent on covering their tracks.